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Subject: Nikkei 225 (JPY225) & Dow Jones Industrial Average (US30)

Period: July 2023 - December 2025

Period Written: September 2025

**Macroeconomic Influences on Equity Indices: A Quantitative Analysis of JPY225 and
US30 Performance**

Executive Summary

This report analyzes the 24-month performance of Japan's JPY225 (Nikkei 225) and the American US30 (Dow Jones Industrial Average), identifying key drivers and the impact of macroeconomic policies, corporate reforms and geopolitical developments. As benchmarks of national financial performance, the Nikkei 225 and US30 serve as primary indicators of market performance in Japan and the United States, and can be used to reflect investor sentiment and economic conditions utilizing a curated list of leading companies.

Market Performance Review Timeline

Both the Nikkei 225 (JPY225) and the Dow Jones Industrial Average (US30) had significant growth over the past two years, due to global economic resilience and strong performing industries. Below is a detailed review of their market performance.

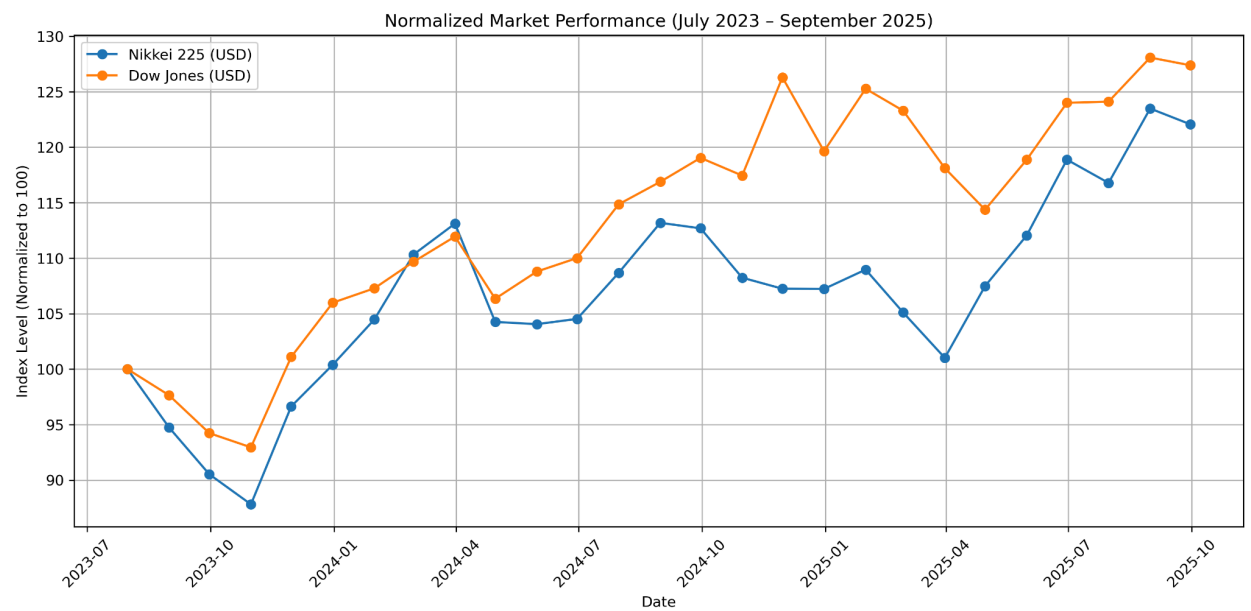


Figure 1.1

Figure 1 is a visual representation of the market performance of the Nikkei 225 and the Dow Jones, with an initial normalization at 100, in order to provide a baseline for relative performance comparison.

The initial baseline in July 2023 is when both the Japanese and American economies are managing high global interest rates and the residual impacts of COVID-19. The United States has had several aggressive post-pandemic interest rate hikes, with a rate increase in both March,

May and July of 2023 lifting rates into the 5 percent range, while Japan (BOJ) continued to maintain its negative interest rate into 2024.

In the latter end of 2023 both indexes experienced a decline, though the Nikkei 225 recorded a steeper drop when measured in USD. This relative underperformance is largely caused by a depreciating yen, which itself is driven by the interest rate policy divergence between the Federal Reserve and the Bank of Japan. (It is important to note that currency fluctuations can skew cross-market comparisons, which is shown here through the Nikkei 225's performance being weaker in USD than its domestic currency).

Starting in December of 2023 and continuing through April 2024 both indices saw a strong recovery. The rebound reflected both global disinflationary trends, as well as the stabilization of energy prices after years of volatility associated with the pandemic and war in Ukraine. As energy costs had been a major cause of global inflation, this stabilization eased pressures and improved investor sentiment with regards to inflation cooling. In the United States, this fueled optimism that the Federal Reserve would begin cutting rates in 2024. In Japan, momentum was boosted by strong export performance, aided by a weak yen accompanied by renewed investor confidence from foreign capital inflow and corporate governance reforms. Notably, Japan ended its negative rate policy by increasing interest rates in March 2024, the first hike in 17 years. While modest compared to the United States, the move is a sign of Japan's growing economic resilience. During this period the two indices tracked each other closely, reflecting global synchronization.

After peaking in April 2024, both indices went through a correction and while the Dow Jones regained momentum the Nikkei 225 stalled. Starting in mid-2024, the Dow Jones entered a steady upward trend, attributed to strong United States earnings, investor confidence in a soft

landing and continued currency strength. In contrast the Nikkei 225 traded largely sideways, with a weaker yen undermining USD-denominated returns despite resilient domestic gains. Early 2025 brought volatility to both indices, but the Dow Jones rebounded by midyear whereas the Nikkei 225's recovery lagged. This created a distinct performance gap in which the Dow Jones' growth outpaced the Nikkei 225's, driven by both stronger market fundamentals and favorable currency dynamics.

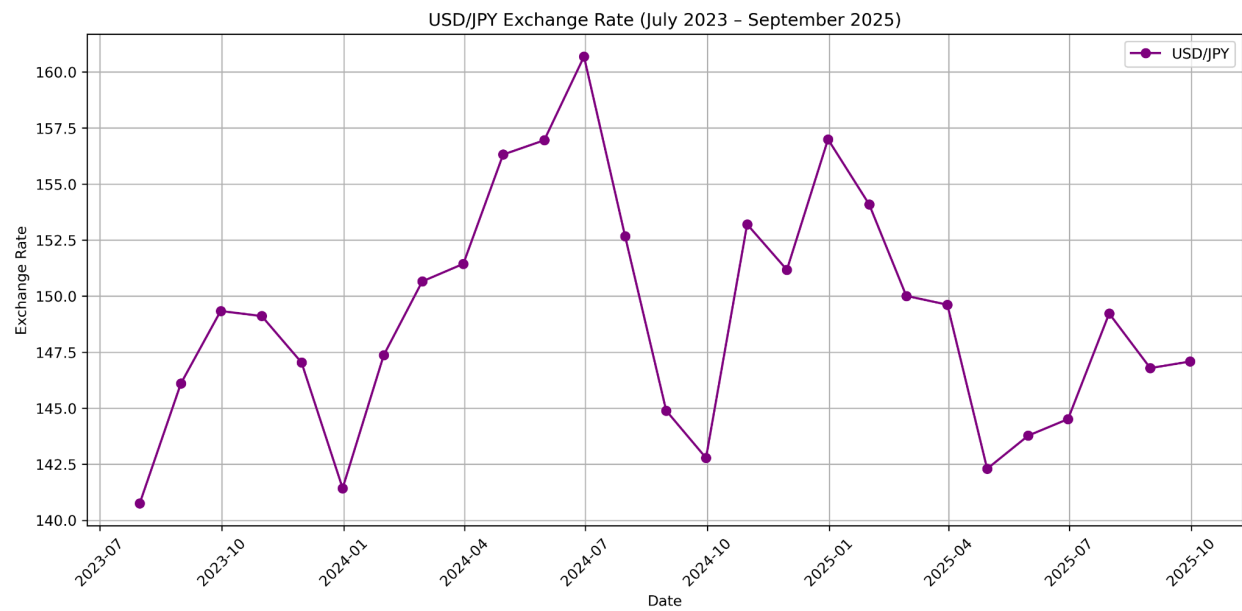


Figure 2.1

The performance of the yen relative to the dollar contributes to the trends shown in Figure 1.1, complemented by Figure 2.1.

Market Performance Correlation Analysis

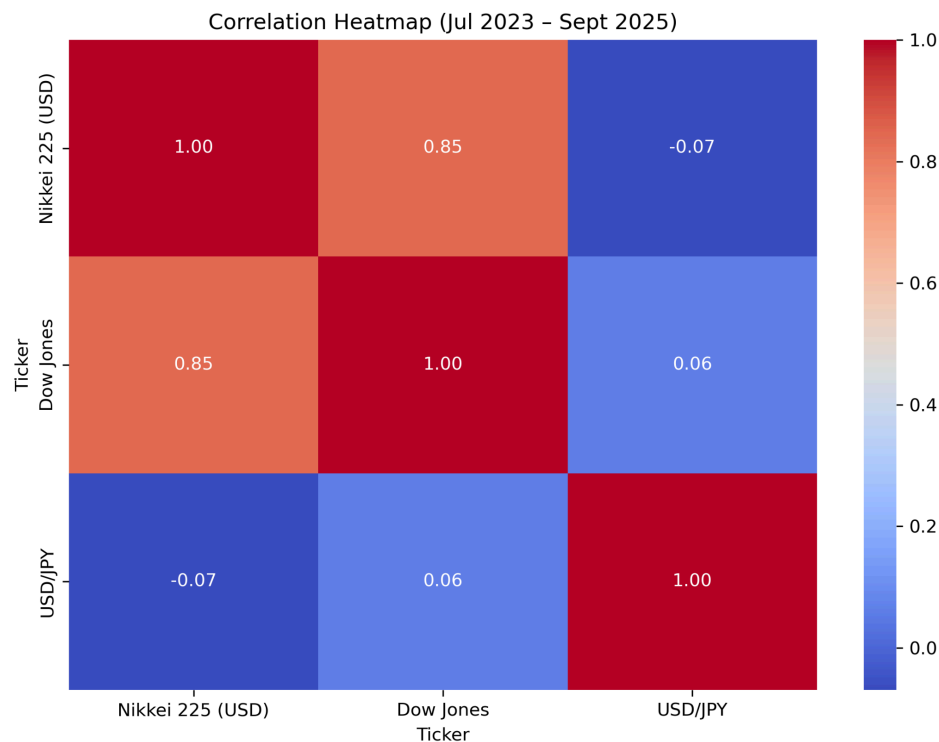


Figure 3.1

Figure 3.1 provides further statistical insight using a correlation heatmap of the Nikkei 225, Dow Jones and USD/JPY exchange rate between July 2023 and July 2025. The Nikkei 225 and Dow Jones exhibit a strong positive correlation at 0.82, confirming that the two markets generally behaved similarly, especially in response to common global drivers including disinflation trends, global interest hikes and investor sentiment. It is important to note that the correlation is not perfect, which is consistent with the divergence after mid-2024 where the Dow Jones rose and the Nikkei 225 stalled/fell.

Notably, the correlation between the Nikkei 225 (USD) and USD/JPY is -0.01. This further emphasizes how the exchange rate effect subdues the performance of Japanese equities when measured in USD. While the depreciating yen supported exports and local equity gains, it also

reduced USD-denominated returns. Contrastingly, the Dow Jones shows a weak positive correlation with USD/JPY at 0.11, suggesting that U.S. equity performance is significantly affected by domestic fundamental performance as opposed to exchange rate fluctuations.

These findings reinforce the earlier analysis, while global macroeconomic forces kept the Nikkei 225 and Dow Jones broadly in sync, currency movements proved decisive in amplifying the performance gap between the indices.

Sector Cumulative Returns

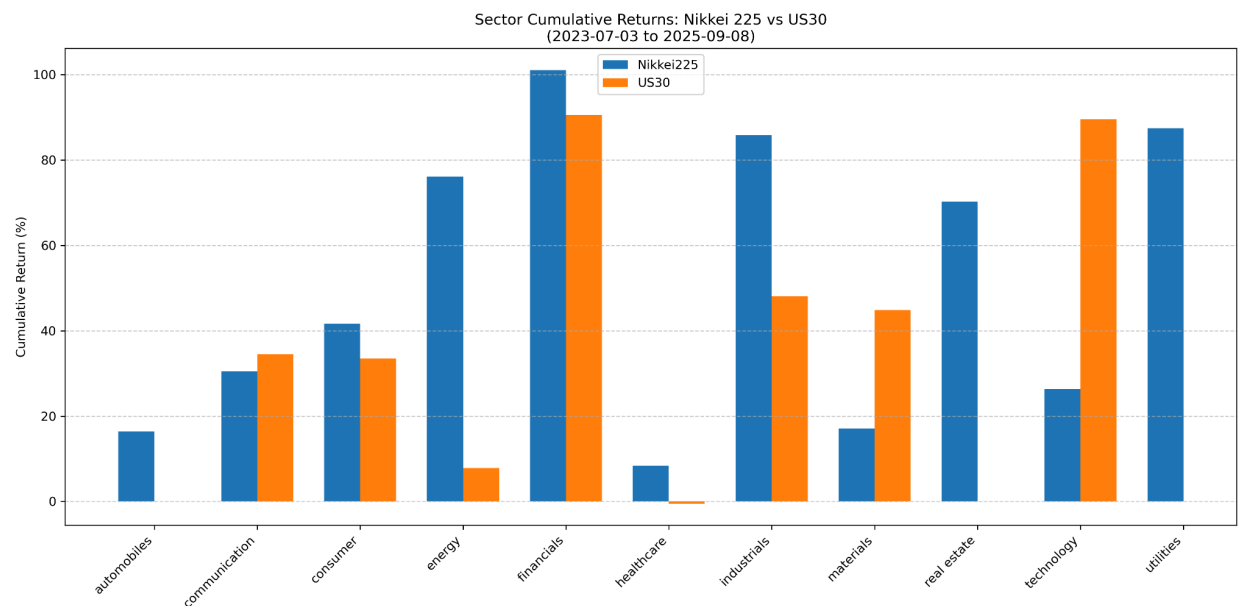


Figure 4.1

Figure 4.1 is a chart that breaks down performance drivers within both indices, moving from a macro level perspective to the micro level.

The Dow Jones’ growth was driven by information technology and financials. The technology sector grew with AI fueled growth, including machine learning models being integrated into

most large scale company processes and continued stable earnings. The financial sector saw growth with banks and insurers benefitting from wider interest rate margins due to the rate hikes. For the Nikkei 225 performance was driven by a larger combination of sectors, the data shows performance drivers in financials, utilities, industrials, energy, and real estate. Financials benefited from global rates rising, and the Bank of Japan did raise rates for the first time, utilities and energy grew from elevated commodity prices, industrials and real estate were stabilized by strong capital inflows. Contrastingly, traditional export leaders like Automobiles and Technology relatively underperformed, indicating supply chain constraints and global demand changes offset potential currency advantages.

Valuation and Market Sentiment Analysis

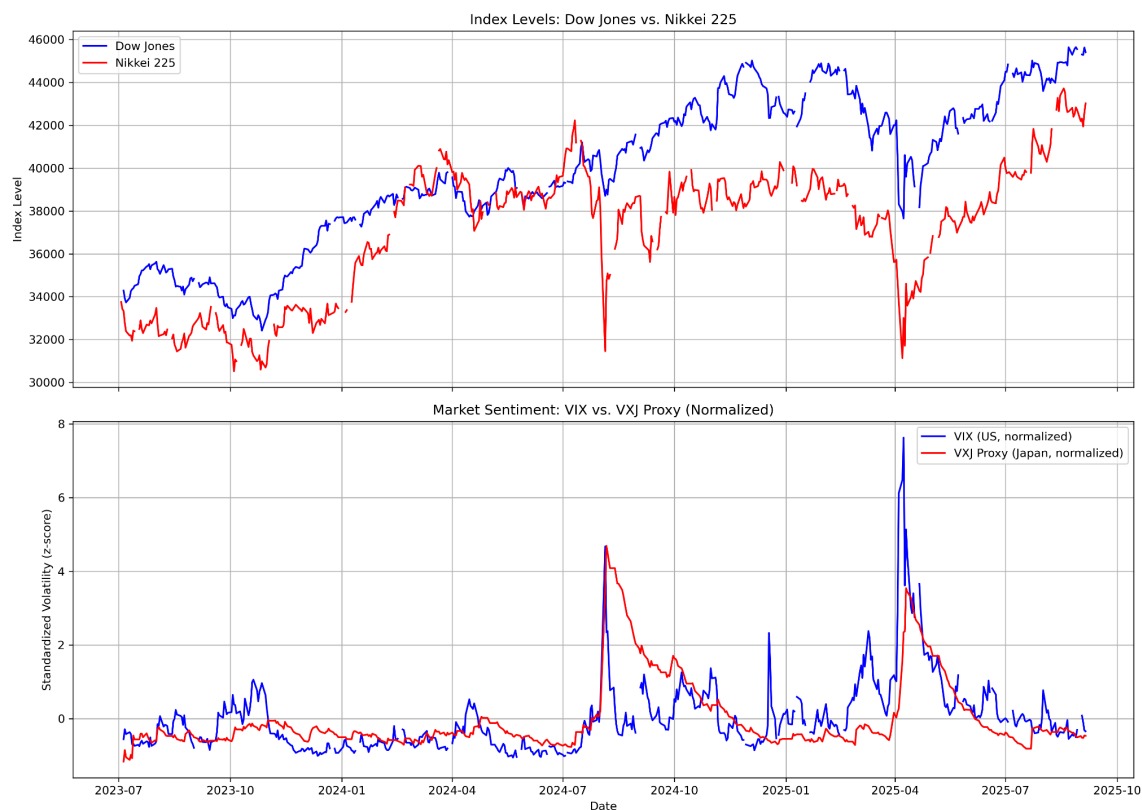


Figure 5.1

Figure 5.1 shows the valuation and market sentiment for both the Nikkei 225 and Dow Jones. The figure takes into account both normalized index levels for both indices along with standardized measures of market volatility, using the VIX and a proxy for the VXJ (to keep consistency with sources for information and maximize accuracy). The volatility measures show when and how sentiment shocks corresponded with market movements.

Starting in late 2023 continued through mid-2024 both the VIX and VXJ remained relatively stable coinciding with broad recovery in equity indices showing optimism as the indices grew and stabilized.

In mid-2024 a sharp rise in the VXJ took place while a drop in the Nikkei 225 happened at the same time, highlighting Japan-specific issues stemming from currency dynamics and policy uncertainty. At the same time the VIX also experienced a rise complimented by only a small decrease in the Dow Jones, but the VIX quickly self-corrected while the VXJ took longer to decrease, emphasizing that U.S. equities were less affected by these developments.

In early 2025 both volatility indices spiked with corresponding declines in the equity indices.

Tariff Policy Shocks and Market Responses

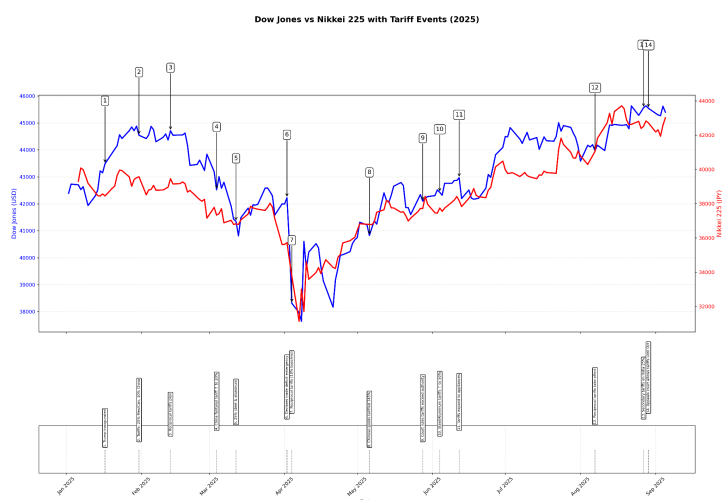


Figure 6.1

Figure 6.1 overlays the performance of both the Dow Jones and Nikkei 225 with a timeline of major U.S. and international tariff policy, with a focus on January to September 2025. This figure highlights how protective trade measures and retaliatory actions generate visible volatility in both indices.

In early 2025, volatility corresponded with initial tariffs following president Trump's inauguration, initiating moderate downward pressure on both indices. The largest market adjustment took place in April, when the trade deficit and reciprocal tariffs were announced. Both indices saw a sharp decline at this time, with the Nikkei 225 stabilizing faster than the Dow Jones. Subsequent events all show a decline for the Dow Jones in particular coinciding with renewed volatility, reflecting the impact of trade uncertainty in the market.

6-month Potential Outlook using Monte Carlo Simulation

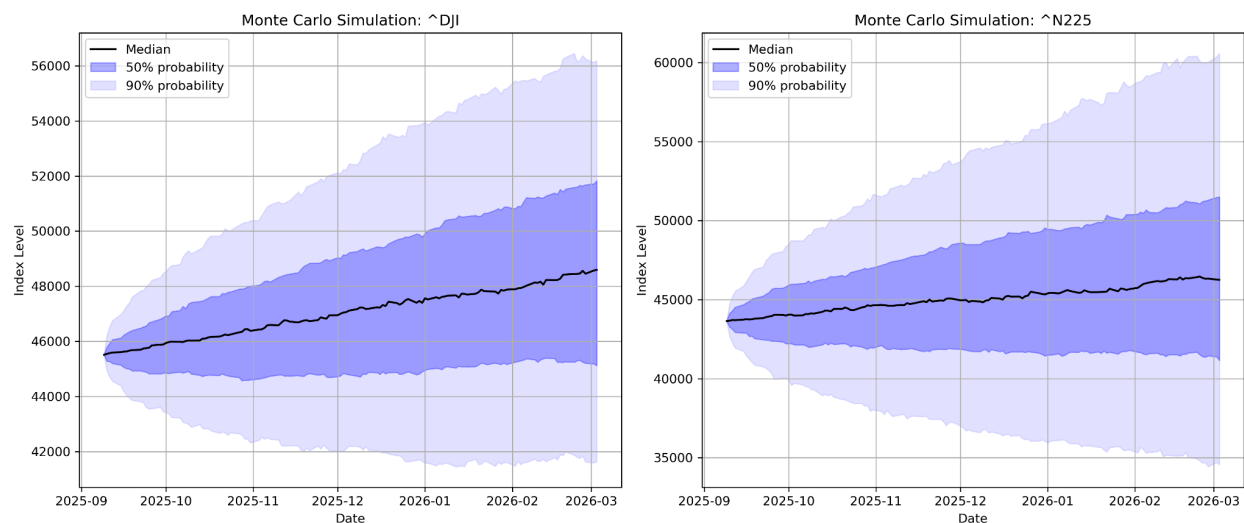


Figure 7.1

The Monte Carlo fan charts show a probabilistic forecast of both index levels over a 6-month horizon starting in September 2025.

Both indices show a positive median trajectory, reflecting historical behaviours. The Dow Jones shows higher median level growth consistent with previous performance. The darker blue probability bands represent the 50% range, and indicate where most simulated outcomes take place, the Dow Jones' band is narrow relative to the Nikkei 225 indicating lower near-term volatility. The lighter blue bands show extreme outcomes representing the 90% probability range, where both indices exhibit asymmetry, with potential upside being larger than downside risk for this region. The wider spread for the Nikkei 225 indicates greater relative uncertainty, which may be partly due to currency risk, the Dow shows a more concentrated band indicating comparatively higher confidence in the 6-month term.

The simulations indicate that both indices are expected to see modest growth over the next 6 months.

Appendix

Figure 1.1 - Normalized Market Performance

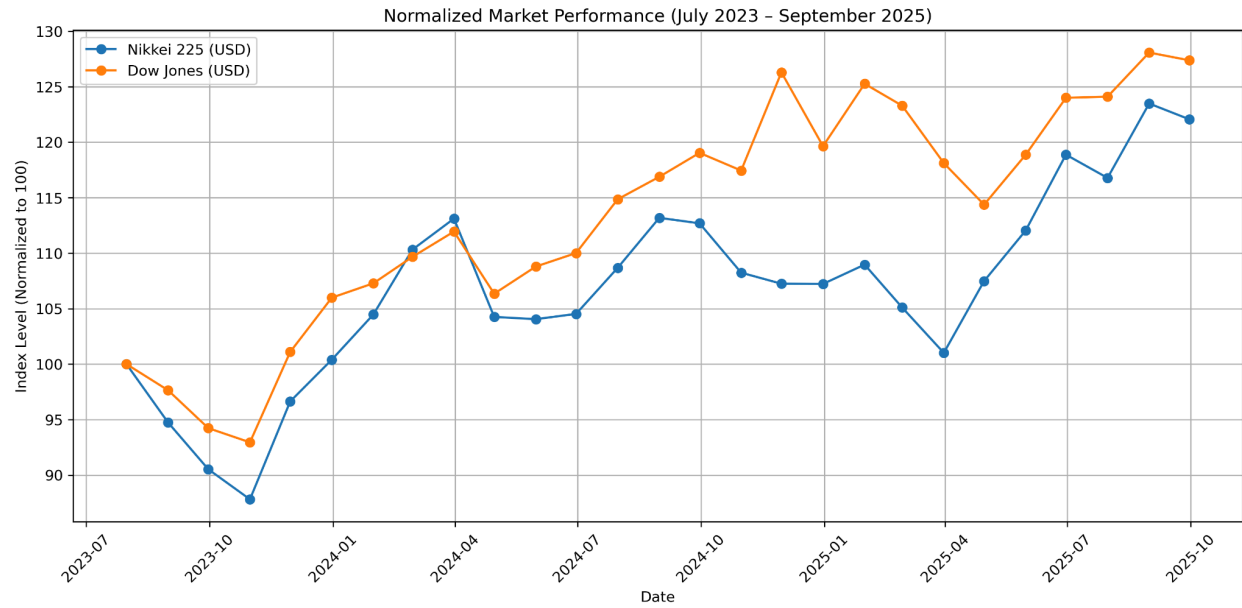


Figure 1.2 - Code for figure 1.1

```
normalized = monthly[["Nikkei 225 (USD)", "Dow Jones"]] / monthly[["Nikkei 225 (USD)", "Dow Jones"]].iloc[0] * 100

plt.figure(figsize=(12,6))
plt.plot(normalized.index, normalized["Nikkei 225 (USD)"], label="Nikkei 225 (USD)", marker="o")
plt.plot(normalized.index, normalized["Dow Jones"], label="Dow Jones (USD)", marker="o")
plt.title("Normalized Market Performance (July 2023 - September 2025)")
plt.xlabel("Date")
plt.ylabel("Index Level (Normalized to 100)")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("graph1_normalized.png", dpi=300, bbox_inches="tight")
plt.show()
```

Figure 2.1 - USD/JPY Exchange Rate

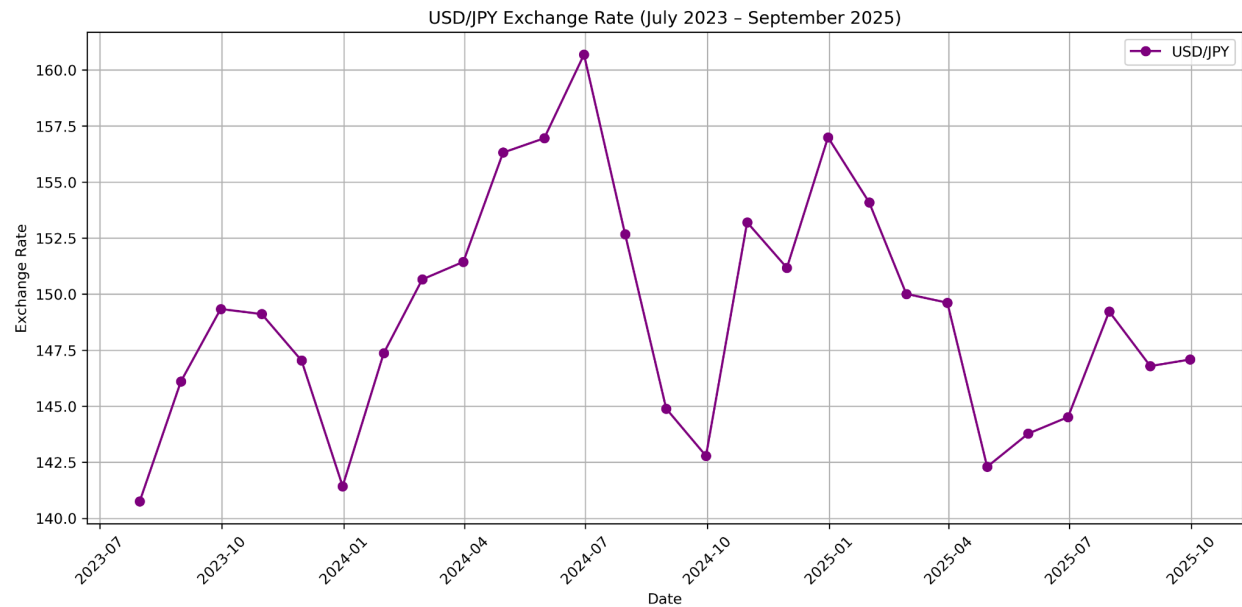


Figure 2.2 - Code for Figure 2.1

```
plt.figure(figsize=(12,6))
plt.plot(monthly.index, monthly["USD/JPY"], label="USD/JPY", color="purple", marker="o")
plt.title("USD/JPY Exchange Rate (July 2023 - September 2025)")
plt.xlabel("Date")
plt.ylabel("Exchange Rate")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("graph2_usdjpy.png", dpi=300, bbox_inches="tight")
plt.show()
```

Figure 3.1 - Correlation Heatmap

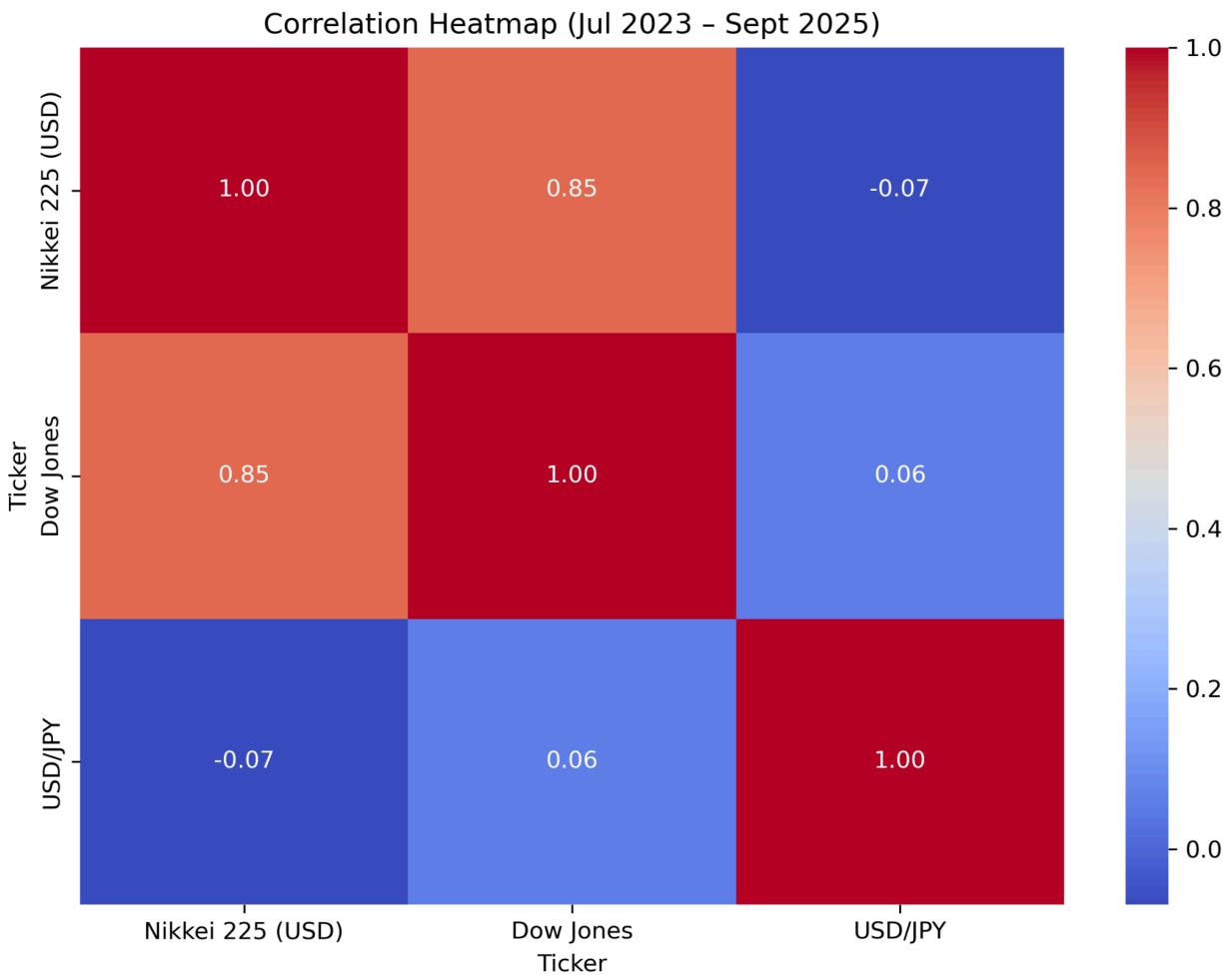


Figure 3.2 - Code for Figure 3.1

```
plt.figure(figsize=(8,6))
corr = monthly[["Nikkei 225 (USD)", "Dow Jones", "USD/JPY"]].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap (Jul 2023 - Sept 2025)")
plt.tight_layout()
plt.savefig("graph3_heatmap.png", dpi=300, bbox_inches="tight")
plt.show()
```

Figure 4.1 - Sector Cumulative Returns

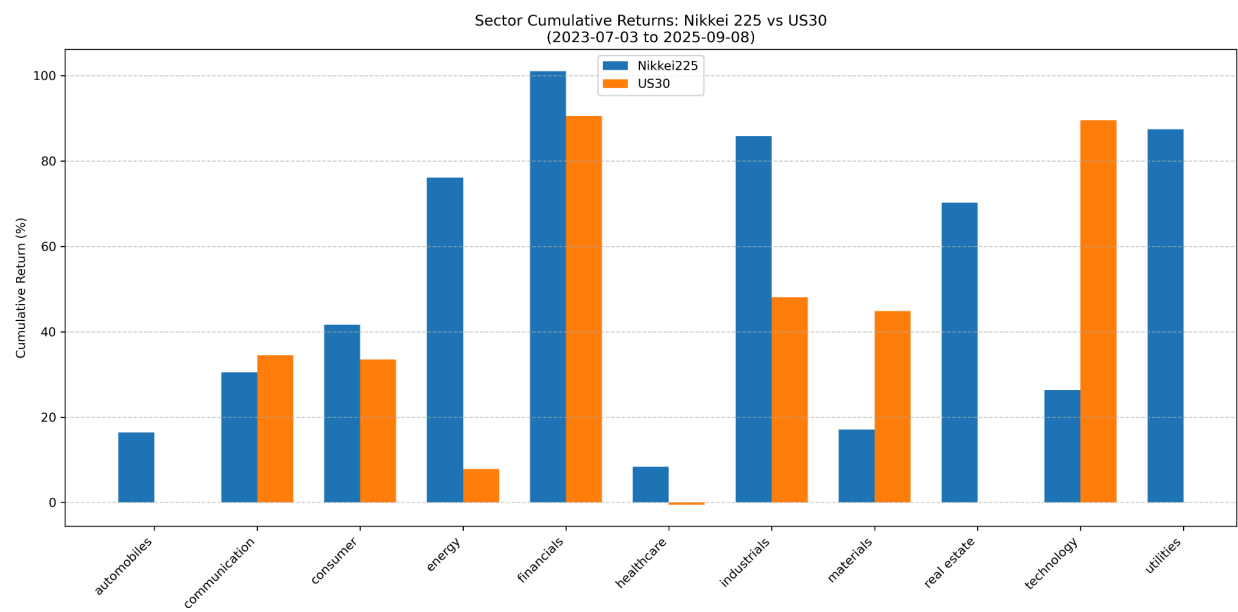


Figure 4.2, 4.3, 4.4 - Code for Figure 4.1

```
import yfinance as yf
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

nikkei_df = pd.read_csv("nikkei225_stocks.csv")
nikkei_df["Index"] = "Nikkei225"

us30_df = pd.read_csv("us30_stocks.csv")
us30_df["Index"] = "US30"

all_stocks = pd.concat([nikkei_df, us30_df], ignore_index=True)

def normalize_sector(name):
    if pd.isna(name):
        return "unknown"
    name = name.lower().strip()
    mapping = {
        # Finance
        "banks": "financials",
        "financial services": "financials",
        "insurance": "financials",
        # Consumer
        "consumer staples": "consumer",
        "consumer goods": "consumer",
        "consumer discretionary": "consumer",
        "retail": "consumer",
        "food products": "consumer",
        "beverages": "consumer",
        "entertainment": "consumer",
        # Technology
        "information technology": "technology",
        "electronic equipment": "technology",
        "semiconductors": "technology",
        "technology": "technology",
        # Industrials
        "industrial machinery": "industrials",
        "construction": "industrials",
        "transportation": "industrials",
        "trading companies": "industrials",
        "building materials": "industrials",
        "airlines": "industrials",
        # Materials
        "chemicals": "materials",
        "steel": "materials",
        "paper & pulp": "materials",
        # Healthcare
        "pharmaceuticals": "healthcare",
        "healthcare equipment": "healthcare",
        # Energy & Utilities
        "energy": "energy",
        "utilities": "utilities",
```



```

        # Real Estate
        "real estate": "real estate",
        # Communication
        "communication services": "communication",
        "telecommunications": "communication",
        "advertising": "communication",
        # Security
        "security services": "industrials", # group under industrials
    }
    return mapping.get(name, name)

all_stocks["Sector"] = all_stocks["Sector"].apply(normalize_sector)

unknown_sectors = all_stocks[all_stocks["Sector"] == "unknown"]
if not unknown_sectors.empty:
    print("Tickers with unknown sector:\n", unknown_sectors[["Ticker", "Index"]])

all_tickers = all_stocks["Ticker"].tolist()
print("Downloading price data for all tickers...")
price_data = yf.download(all_tickers, start="2023-07-01", end="2025-09-09")["Close"]

# Keep track of actual range
start_date = price_data.index[0].strftime("%Y-%m-%d")
end_date = price_data.index[-1].strftime("%Y-%m-%d")

normalized = price_data / price_data.iloc[0] * 100

sector_growth = {}
for idx in all_stocks["Index"].unique():
    df_index = all_stocks[all_stocks["Index"] == idx]
    sector_df = pd.DataFrame(index=normalized.index)

    for sector in df_index["Sector"].unique():
        if sector != "unknown":
            tickers_in_sector = df_index[df_index["Sector"] == sector]["Ticker"].tolist()
            sector_df[sector] = normalized[tickers_in_sector].mean(axis=1)

    sector_growth[idx] = sector_df

all_sectors = sorted(set(sector_growth["Nikkei225"].columns).union(sector_growth["US30"].columns))

cum_returns = pd.DataFrame({
    "Sector": all_sectors,
    "Nikkei225": [sector_growth["Nikkei225"][s].iloc[-1] - 100 if s in sector_growth["Nikkei225"].columns else np.nan for s in all_sectors],
    "US30": [sector_growth["US30"][s].iloc[-1] - 100 if s in sector_growth["US30"].columns else np.nan for s in all_sectors]
})

```

```

cum_returns.to_csv("sector_cumulative_returns.csv", index=False)
print("Cumulative returns saved to sector_cumulative_returns.csv.")

x = np.arange(len(cum_returns["Sector"]))
width = 0.35

fig, ax = plt.subplots(figsize=(14, 7))

rects1 = ax.bar(x - width/2, cum_returns["Nikkei225"], width, label='Nikkei225', color="#1f77b4")
rects2 = ax.bar(x + width/2, cum_returns["US30"], width, label='US30', color="#ff7f0e")

ax.set_ylabel('Cumulative Return (%)')
ax.set_title(f'Sector Cumulative Returns: Nikkei 225 vs US30\n({start_date} to {end_date})')
ax.set_xticks(x)
ax.set_xticklabels(cum_returns["Sector"], rotation=45, ha='right')
ax.legend()

ax.grid(True, axis='y', linestyle='--', alpha=0.6)
plt.tight_layout()
plt.savefig("sector_cumulative_returns.png", dpi=300)
plt.show()

```

Figure 5.1 - Valuation Indicator

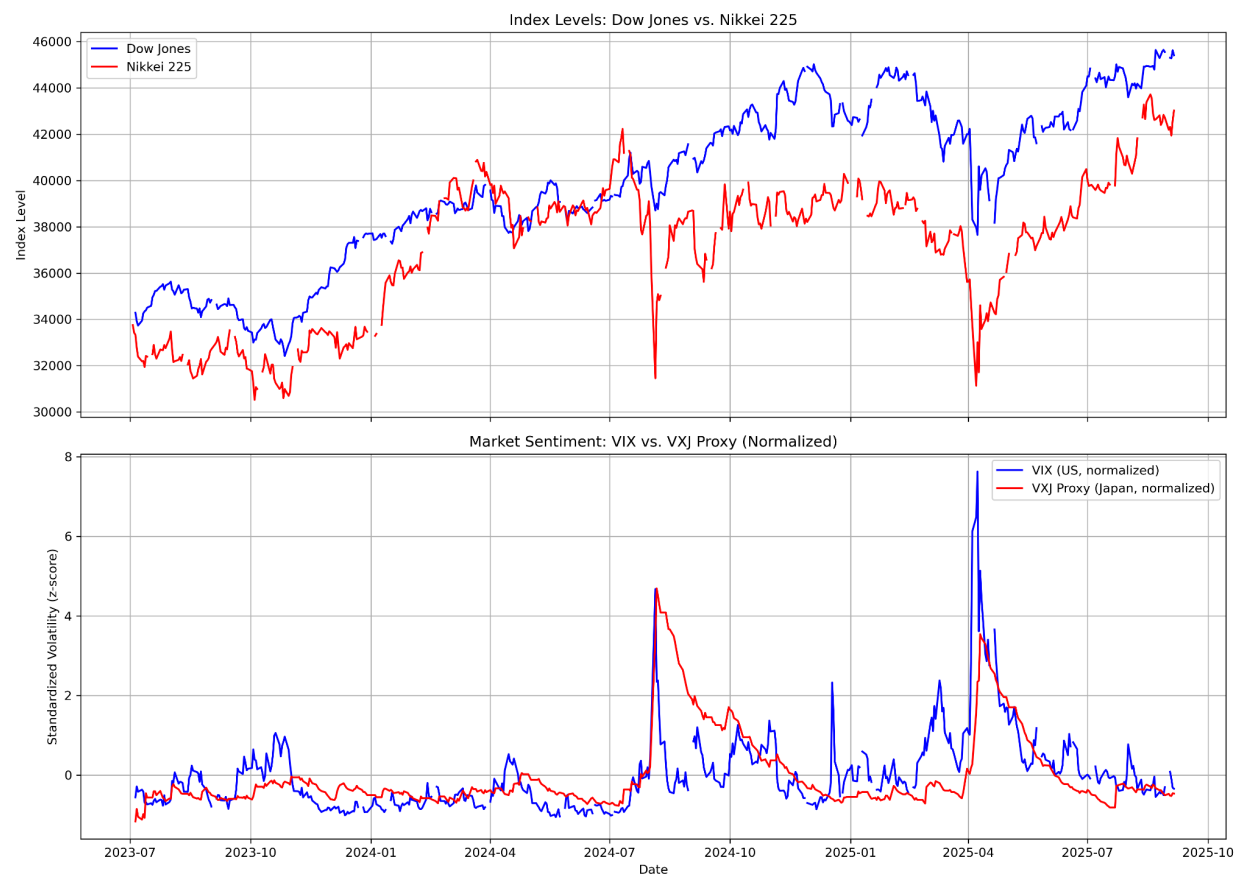


Figure 5.2 - Code for Figure 5.1

```
tickers = ["^DJI", "^N225", "^VIX"] # Dow, Nikkei, US VIX
start_date = "2023-07-01"
end_date = "2025-09-08"

data = yf.download(tickers, start=start_date, end=end_date)["Close"]

data = data.rename(columns={
    "^DJI": "Dow Jones",
    "^N225": "Nikkei 225",
    "^VIX": "VIX"
})

nikkei_log_returns = np.log1p(data["Nikkei 225"].pct_change(fill_method=None)) # ✓ fixed
vxj_proxy = nikkei_log_returns.ewm(span=30).std() * (252**0.5) * 100 # annualized %

# Optional: normalize both series for comparability
vix_norm = (data["VIX"] - data["VIX"].mean()) / data["VIX"].std()
vxj_norm = (vxj_proxy - vxj_proxy.mean()) / vxj_proxy.std()

fig, axes = plt.subplots(2, 1, figsize=(14, 10), sharex=True)

axes[0].plot(data.index, data["Dow Jones"], label="Dow Jones", color="blue")
axes[0].plot(data.index, data["Nikkei 225"], label="Nikkei 225", color="red")
axes[0].set_ylabel("Index Level")
axes[0].set_title("Index Levels: Dow Jones vs. Nikkei 225")
axes[0].legend()
axes[0].grid(True)

axes[1].plot(data.index, vix_norm, label="VIX (US, normalized)", color="blue")
axes[1].plot(vxj_norm.index, vxj_norm, label="VXJ Proxy (Japan, normalized)", color="red")
axes[1].set_ylabel("Standardized Volatility (z-score)")
axes[1].set_title("Market Sentiment: VIX vs. VXJ Proxy (Normalized)")
axes[1].legend()
axes[1].grid(True)

plt.xlabel("Date")
plt.tight_layout()
plt.savefig("vals2.png", dpi=300)
plt.show()
```

Figure 6.1 - Tariff Event Timeline

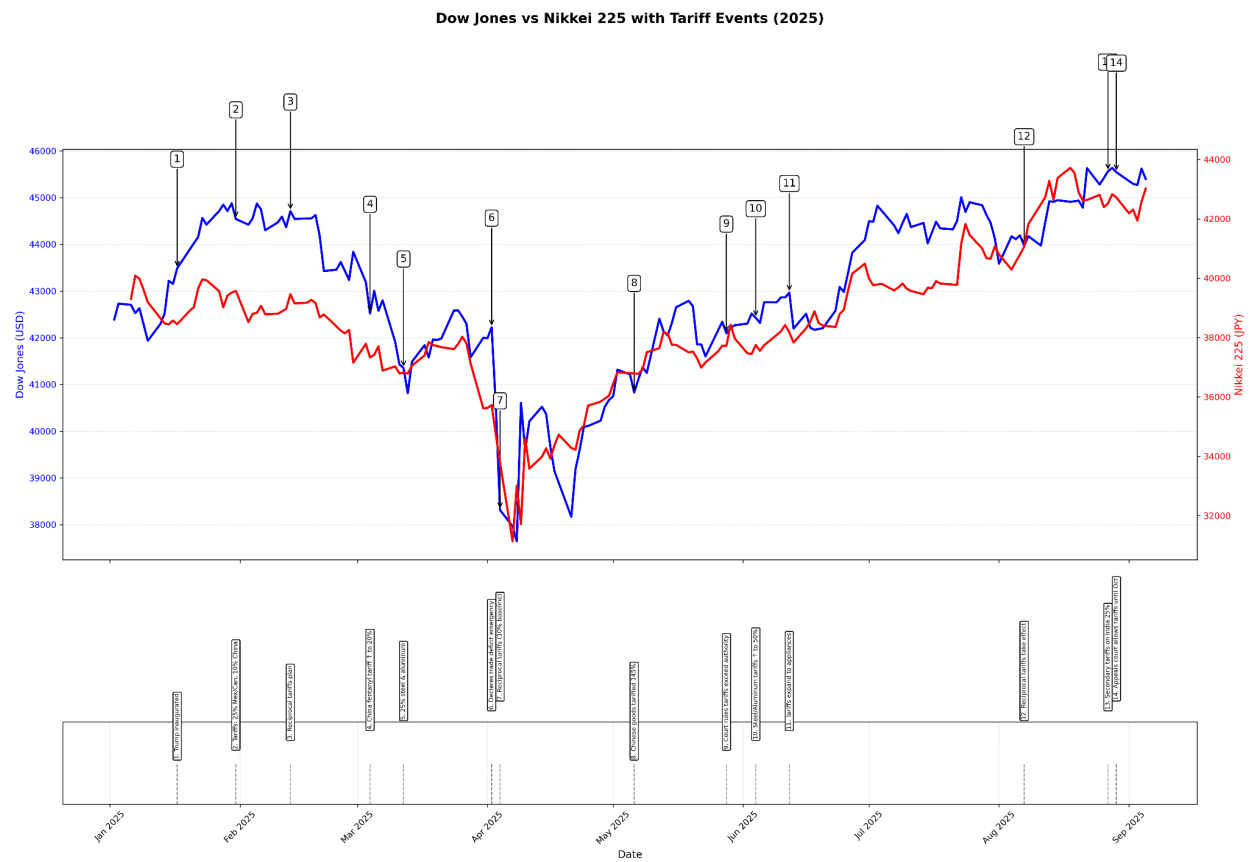


Figure 6.2, 6.3, 6.4 - Code for Figure 6.1

```
import yfinance as yf
import pandas as pd
import matplotlib.pyplot as plt

# Download index data
nikkei = yf.download("^N225", start="2025-01-01", end="2025-09-08")["Close"].ffill()
dow = yf.download("^DJI", start="2025-01-01", end="2025-09-08")["Close"].ffill()

# Confirmed events only sourced from Wikipedia
events = [
    ("2025-01-20", "Trump inaugurated"),
    ("2025-02-01", "Tariffs: 25% Mex/Can, 10% China"),
    ("2025-02-13", "Reciprocal tariffs plan"),
    ("2025-03-04", "China fentanyl tariff ↑ to 20%"),
    ("2025-03-12", "25% steel & aluminum"),
    ("2025-04-02", "Declares trade deficit emergency"),
    ("2025-04-05", "Reciprocal tariffs (10% baseline)"),
    ("2025-05-06", "Chinese goods tariffed 145%"),
    ("2025-05-28", "Court rules tariffs exceed authority"),
    ("2025-06-04", "Steel/Aluminum tariffs ↑ to 50%"),
    ("2025-06-12", "Tariffs expand to appliances"),
    ("2025-08-07", "Reciprocal tariffs take effect"),
    ("2025-08-27", "Secondary tariffs on India 25%"),
    ("2025-08-29", "Appeals court allows tariffs until Oct")
]

events = sorted([(pd.Timestamp(date), label) for date, label in events], key=lambda x: x[0])

# Create figure
fig, (ax1, ax2) = plt.subplots(
    2, 1, figsize=(20, 14), gridspec_kw={'height_ratios': [5, 1]}, sharex=True
)

# Plot Dow
ax1.plot(dow.index, dow, label="Dow Jones", color="blue", linewidth=2.5)
ax1.set_ylabel("Dow Jones (USD)", color="blue", fontsize=12)
ax1.tick_params(axis='y', labelcolor="blue")
ax1.grid(True, which='major', axis='y', linestyle='--', alpha=0.3)

# Plot Nikkei
ax3 = ax1.twinx()
ax3.plot(nikkei.index, nikkei, label="Nikkei 225", color="red", linewidth=2.5)
ax3.set_ylabel("Nikkei 225 (JPY)", color="red", fontsize=12)
ax3.tick_params(axis='y', labelcolor="red")
```

```

# Annotate events
arrow_color = "black"
arrow_offset = 120

for i, (date, label) in enumerate(events):
    if date not in dow.index:
        nearest_idx = dow.index.get_indexer([date], method='ffill')[0]
        date = dow.index[nearest_idx]

    y = dow.loc[date]
    ax = ax1

    ax.annotate(
        str(i+1),
        xy=(date, y),
        xytext=(0, arrow_offset),
        textcoords="offset points",
        arrowprops=dict(arrowstyle="->", color=arrow_color, lw=1.2),
        fontsize=12,
        color=arrow_color,
        ha="center",
        va="bottom",
        bbox=dict(facecolor='white', edgecolor=arrow_color, boxstyle='round,pad=0.3')
    )

# Timeline panel
ax2.set_ylim(0, 1)
ax2.set_yticks([])
ax2.set_xlabel("Date", fontsize=12)
ax2.grid(True, axis='x', linestyle='--', alpha=0.3)

```

```

for i, (date, label) in enumerate(events):
    if date not in dow.index:
        nearest_idx = dow.index.get_indexer([date], method='ffill')[0]
        date = dow.index[nearest_idx]

    ax2.plot([date, date], [0, 0.5], color='gray', linestyle='--', lw=1)

    stagger = 0.05 + (i % 7) * 0.12
    text_y = 0.5 + stagger

    ax2.text(
        date, text_y,
        f"{i+1}. {label}",
        rotation=90,
        fontsize=7,
        ha='center',
        va='bottom',
        color='black',
        bbox=dict(facecolor='white', edgecolor='black', boxstyle='round,pad=0.2')
    )

# Format x-axis
ax2.set_xticks(pd.date_range(start="2025-01-01", end="2025-09-01", freq='MS'))
ax2.set_xticklabels(
    [d.strftime('%b %Y') for d in pd.date_range(start="2025-01-01", end="2025-09-01", freq='MS')],
    rotation=45
)

# Title
fig.suptitle(
    "Dow Jones vs Nikkei 225 with Tariff Events (2025)",
    fontsize=16, fontweight='bold'
)

plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.savefig("tariff_timeline_confirmed.png", dpi=300)
plt.show()

```

Figure 7.1 - Monte Carlo Simulation

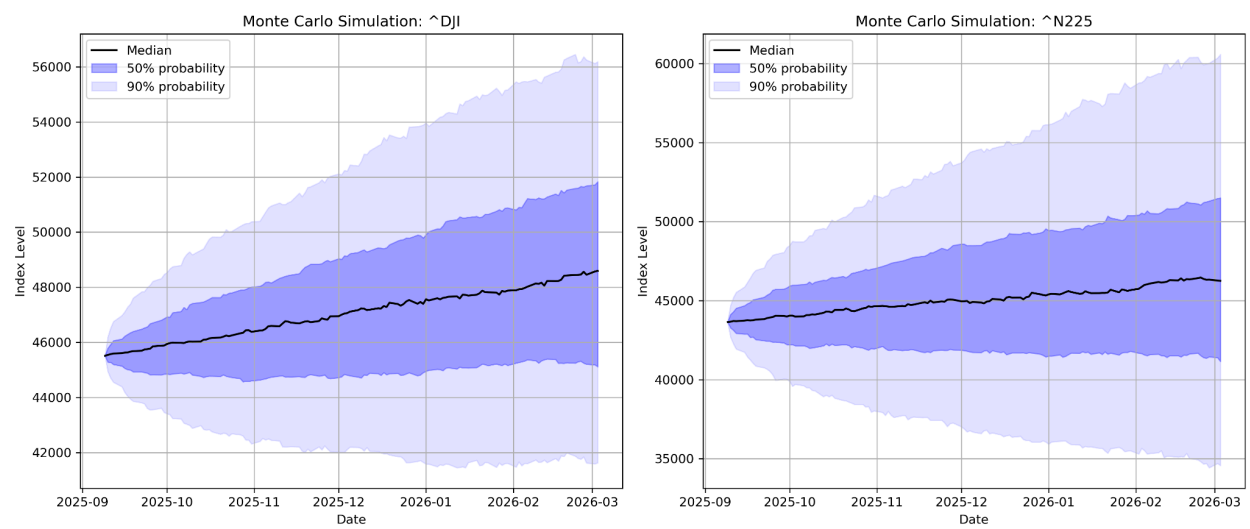


Figure 7.2, 7.3 - Code for Figure 7.1

```
tickers = ["^N225", "^DJI"]
start_date = "2023-07-01"
end_date = "2025-09-09"

raw_data = yf.download(tickers, start=start_date, end=end_date, auto_adjust=True)

if isinstance(raw_data.columns, pd.MultiIndex):
    data = raw_data['Close']
else:
    data = raw_data # single-level (unlikely for multiple tickers)

print("Columns used for simulation:", data.columns)
print(data.head())

returns = data.pct_change().dropna()
vol = returns.std() * np.sqrt(252) # annualized volatility
mu = returns.mean() * 252         # annualized drift

n_days = 126 # ~6 months
dt = 1/252
n_sim = 1000
np.random.seed(42)

def simulate_paths(S0, mu, sigma, n_days, n_sim):
    paths = np.zeros((n_days, n_sim))
    paths[0] = S0
    for t in range(1, n_days):
        z = np.random.standard_normal(n_sim)
        paths[t] = paths[t-1] * np.exp((mu - 0.5*sigma**2)*dt + sigma*np.sqrt(dt)*z)
    return paths

simulations = {}
for ticker in data.columns:
    S0 = data[ticker].iloc[-1]
    simulations[ticker] = simulate_paths(S0, mu[ticker], vol[ticker], n_days, n_sim)
```

```

plt.figure(figsize=(14,6))

for i, ticker in enumerate(data.columns):
    plt.subplot(1,2,i+1)
    x = pd.date_range(start=data.index[-1]+pd.Timedelta(days=1), periods=n_days, freq='B')

    # compute percentiles
    p5 = np.percentile(simulations[ticker], 5, axis=1)
    p25 = np.percentile(simulations[ticker], 25, axis=1)
    p50 = np.percentile(simulations[ticker], 50, axis=1)
    p75 = np.percentile(simulations[ticker], 75, axis=1)
    p95 = np.percentile(simulations[ticker], 95, axis=1)

    # plot median
    plt.plot(x, p50, color='black', label='Median')

    # fill 50% and 90% bands
    plt.fill_between(x, p25, p75, color='blue', alpha=0.3, label='50% probability')
    plt.fill_between(x, p5, p95, color='blue', alpha=0.1, label='90% probability')

    plt.title(f"Monte Carlo Simulation: {ticker}")
    plt.xlabel("Date")
    plt.ylabel("Index Level")
    plt.legend()
    plt.grid(True)

plt.tight_layout()
plt.savefig("monte_carlo_fan_chart.png", dpi=300, bbox_inches="tight")
plt.show()

```